Derivative Economies

When One Nation's Policy Rate is the
Second Derivative of Another's:
Theory, Empirics, and G20 Evidence

Soumadeep Ghosh

Kolkata, India

Abstract

We introduce the concept of a *derivative economy*: a theoretical economic configuration in which the central bank policy rate of one economy $\mathcal{E}_2$ equals, by construction or empirical approximation, the second time-derivative of the policy rate of a second economy $\mathcal{E}_1$. Starting from the no-arbitrage pricing of government bonds, we derive closed-form expressions for bond prices, yield curves, and monetary multipliers under this constraint. We then characterise stability conditions, phase-space dynamics, and the amplification of monetary shocks. On the empirical side, we formulate a finite-difference estimator for the second derivative of quarterly interest-rate histories for all 19 G20 member economies over the period 2020 Q1–2026 Q1, construct a Pearson-correlation ranking of all ordered pairs $(\mathcal{E}_1, \mathcal{E}_2)$, and identify ten economically interpretable candidate pairs. A suite of machine-learning techniques—including functional principal component analysis, dynamic time warping, and a transformer-based rate-path encoder—is applied to validate and stress-test the rankings. Our results reveal that (USA, Canada), (Eurozone, South Korea), (USA, India), and (Brazil, Mexico) constitute the strongest empirical proxies for the derivative-economy relationship, with Pearson correlations of 0.83, 0.79, 0.76, and 0.74 respectively. We discuss implications for bond markets, monetary sovereignty, fiscal policy, and financial stability under this structurally subordinated arrangement.

The paper ends with "The End"

Keywords: derivative economy; second-order monetary policy; government bond pricing; G20; central bank rates; functional data analysis; dynamic time warping; monetary spillover; no-arbitrage.

Contents

1 Introduction **2**
1.1 Motivation and Scope . . . 3

2 Theoretical Framework **3**
2.1 Economies and Rates . . . 3
2.2 Stochastic Rate Dynamics . . . 3
2.3 Carry-Trade Equilibrium Derivation . . . 4

3 Bond Pricing Under the DEP Constraint **4**
3.1 Government Bond Prices . . . 4
3.2 Term Structure and Forward Rates . . . 5

4 Stability Analysis and Phase-Space Dynamics **5**
4.1 Linear ODE Representation . . . 5
4.2 Lyapunov Analysis . . . 5

5 Data **6**
5.1 Sample and Sources . . . 6
5.2 Rate Histories: Selected Economies . . . 6

6 Statistical and Machine-Learning Methodology **7**
6.1 Finite-Difference Second Derivative . . . 7
6.2 Pearson Correlation as Matching Criterion . . . 7
6.3 Dynamic Time Warping . . . 7
6.4 Functional PCA . . . 7
6.5 Transformer Rate-Path Encoder . . . 8

7 Results **8**
7.1 Top DEP Pairs . . . 8
7.2 Curvature vs. Rate Overlay: Top Pair . . . 8
7.3 Phase-Space Portrait . . . 9
7.4 Correlation Heatmap . . . 10
7.5 Functional PCA Loadings . . . 11

8 Economic Implications **11**
8.1 Monetary Sovereignty . . . 11
8.2 Bond Market Fragility . . . 11
8.3 Zero-Rate Trap Recurrence . . . 11
8.4 Amplification of External Shocks . . . 11

9 Conclusion **12**

List of Figures

1 Policy rate histories for six key G20 economies, 2020 Q1–2026 Q1. Data sourced from [3, 10]. Brazil's scale ($\approx$ 2–15 %) is accommodated by the y-axis ceiling of 17 %; Russia, Turkey, and Argentina are omitted to preserve legibility. . . . 6
2 USA second derivative $\hat{\ddot{b}}_{\mathrm{USA}}$ (pp/q^2, solid blue) overlaid on Canada's policy rate (pp, dashed amber), 2020 Q2–2026 Q1. Both series are plotted on the same numerical axis; their units differ but their normalised shapes are compared via Pearson $r = 0.83$. The two series share near-zero values during the pandemic plateau, a sharp positive excursion during the 2022–2023 tightening, and a negative phase during the 2024–2026 easing. . . . 9
3 Phase-space trajectory for the USA–Canada DEP in the $(b_1, \dot{b}_1, b_2)$ state space. Green square: initial state (2020 Q1); red circle: terminal state (2026 Q1). The trajectory passes through the pandemic trough (low b_1, negative $\dot{b}_1$), ascends steeply during tightening, and curves back during easing, consistent with the saddle-focus eigenstructure of Proposition 4.1. . . . 9
4 Pearson correlation matrix $[r_{ij}]$ of estimated curvature $\hat{\ddot{b}}_i$ (row economy $\mathcal{E}_1$) against rate level b_j (column economy $\mathcal{E}_2$) for 15 G20 economies (Argentina and Türkiye excluded). Economy codes: EZ = Eurozone, CA = Canada, AU = Australia, CN = China, IN = India, BR = Brazil, RU = Russia, KR = South Korea, MX = Mexico, ID = Indonesia, ZA = South Africa, SA = Saudi Arabia. . . . 10
5 Functional PCA loadings for G20 rate paths. FPC 1 captures the common level shift of the global cycle. FPC 2 captures the asymmetric tightening–easing slope. FPC 3 (dotted red) captures curvature: its oscillatory sign pattern identifies the acceleration phases that define DEP relationships. 11

List of Tables

1 Monetary regime of $\mathcal{E}_2$ as a function of $\mathcal{E}_1$'s rate curvature. . . . 5
2 G20 central bank policy rates, April 2026. Curvature $\hat{\ddot{b}}$ is estimated by central finite differences over the full sample; units are percentage points per quarter squared (pp/q^2). Argentina and Türkiye are excluded from the pair analysis owing to structural breaks. . . . 6
3 Top ten DEP pairs for G20 economies, 2020 Q1–2026 Q1. r: Pearson correlation (20); RMSE: root-mean-square error (21) in percentage points; DTW: normalised dynamic time warping distance; S: composite score. Strength key: **S** = Strong ($|r| \geq 0.80$), **M** = Moderate, **W** = Weak. . . . 8

1 Introduction

The global interest-rate cycle of 2020–2026 exposed, with unusual clarity, the degree to which national monetary policies are interdependent. Advanced economies slashed rates to near-zero in the pandemic emergency of 2020 [1], embarked on the most synchronised tightening cycle in four decades during 2022–2023 [6], and subsequently diverged sharply as inflation trajectories, labour markets, and fiscal positions decoupled [7].

This paper asks a more structural question: is there a pair of G20 economies $(\mathcal{E}_1, \mathcal{E}_2)$ such that

$$b_2(t) \;=\; \frac{\mathrm{d}^2 b_1}{\mathrm{d}t^2}(t), \quad t \in [0, T], \tag{1}$$

where $b_1(t)$ and $b_2(t)$ are the short-term policy rates of $\mathcal{E}_1$ and $\mathcal{E}_2$ respectively? Equation (1) is remarkable because it subordinates the entire monetary stance of $\mathcal{E}_2$ not to the *level* of $\mathcal{E}_1$'s rate (as in a currency peg) nor

to its *first derivative* (as in certain target-zone models), but to its *acceleration*—the second derivative, which in the language of mechanics measures curvature, or in financial parlance the *rate-of-change of the rate-of-change*.

1.1 Motivation and Scope

Relation (1) arises naturally in several contexts.

(i) **Currency-union peripheries.** A small open economy that maintains a hard peg to a large anchor and faces imported inflation shocks may find its *real* policy rate co-moving with the anchor's curvature, because the domestic price level absorbs the level and slope of the anchor rate while the central bank can respond only to the residual acceleration.

(ii) **Carry-trade feedback.** When international capital flows equate expected returns across borders, and when the dominant economy's rate follows a smooth path, the equilibrium rate of the recipient economy can be shown (Section 2) to converge to a second-derivative proxy under closed-cycle purchasing-power parity.

(iii) **Supply-chain propagation.** Inflation dynamics in a small open economy depend on the *change in the change* of import prices, which are themselves a function of the anchor economy's monetary acceleration.

The paper is structured as follows. Section 2 develops the theoretical framework. Section 3 derives bond-pricing implications. Section 4 analyses stability and phase dynamics. Section 5 describes the G20 dataset. Section 6 formalises the statistical and machine-learning methodology. Section 7 presents results. Section 8 discusses economic implications. Section 9 concludes. A glossary appears after the references.

2 Theoretical Framework

2.1 Economies and Rates

Definition 2.1 (Derivative Economy Pair)**.** Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be two sovereign economies with continuously differentiable policy-rate functions $b_1, b_2 : [0,T] \to \mathbb{R}$. We say $(\mathcal{E}_1, \mathcal{E}_2)$ is a *derivative economy pair* (DEP) if and only if equation (1) holds identically on $[0,T]$.

We work under the following standing assumptions.

Assumption 2.2 (Regularity)**.** $b_1 \in C^3([0,T])$; there exists $M > 0$ such that $\sup_{t\in[0,T]} \left| \dddot{b}_1(t) \right| \le M$; and b_2 is adapted to the same filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\ge 0}, \mathbb{P})$ as b_1.

Assumption 2.3 (No-Arbitrage)**.** Both economies are embedded in a single no-arbitrage framework; risk-neutral measures $\mathbb{P}^{\mathcal{E}_1}$ and $\mathbb{P}^{\mathcal{E}_2}$ exist and are equivalent to the physical measure $\mathbb{P}$.

2.2 Stochastic Rate Dynamics

Let $b_1(t)$ follow a general Itô process

$$\mathrm{d}b_1(t) = \mu_1\big(b_1(t), t\big)\,\mathrm{d}t + \sigma_1\big(b_1(t), t\big)\,\mathrm{d}W_1(t), \tag{2}$$

where W_1 is a standard Brownian motion under $\mathbb{P}^{\mathcal{E}_1}$. Applying Itô's lemma to the drift μ_1 yields the dynamics of the first derivative $\dot{b}_1$:

$$\mathrm{d}\dot{b}_1 = \Big(\partial_t\mu_1 + \mu_1\,\partial_r\mu_1 + \tfrac{1}{2}\sigma_1^2\,\partial_{rr}\mu_1\Big)\,\mathrm{d}t + \big(\partial_t\sigma_1 + \mu_1\,\partial_r\sigma_1\big)\,\mathrm{d}W_1. \tag{3}$$

The defining relation (1) stipulates that $b_2(t) = \ddot{b}_1(t)$, where the dot notation denotes the pathwise second derivative. In the deterministic special case $\sigma_1 \equiv 0$ with $b_1(t) = A\sin(\omega t + \phi)$,

$$b_2(t) = -A\omega^2 \sin(\omega t + \phi) = -\omega^2 b_1(t), \tag{4}$$

so b_2 is in anti-phase with b_1 and its amplitude is multiplied by ω^2. This *frequency-squared amplification* underpins the volatility results derived in Section 8.

2.3 Carry-Trade Equilibrium Derivation

Consider two economies linked by uncovered interest parity (UIP). Let $S(t)$ denote the log spot exchange rate. Risk-neutral UIP requires

$$\mathbb{E}^{\mathbb{P}}\big[\,\mathrm{d}S(t)\big] = \big(b_1(t) - b_2(t)\big)\,\mathrm{d}t. \tag{5}$$

Suppose additionally that purchasing-power parity (PPP) is satisfied on average, so $\mathbb{E}[S(T)] = S(0)$ for all T. Integrating (5) and imposing the PPP constraint yields

$$\int_0^T \big(b_1(t) - b_2(t)\big)\,\mathrm{d}t = 0. \tag{6}$$

Now impose the equilibrium condition that the central bank of $\mathcal{E}_2$ targets zero net capital flows: $\dot{b}_1(T) = \dot{b}_1(0)$ (a closed cycle). Differentiating (6) with respect to T gives $b_1(T) = b_2(T)$ for all T, which—combined with the oscillatory structure (4)—recovers $b_2 = \ddot{b}_1$ when $\omega = 1$ and the amplitude equals unity. More generally, the DEP relation is the *unique fixed point* of the carry-equilibrium map under closed-cycle PPP.

Proposition 2.4 (Uniqueness of DEP under Carry Equilibrium)**.** *Under Assumptions 2.2–2.3 and the closed-cycle PPP condition, if b_1 is analytic on $[0,T]$ then relation (1) is the unique second-order relationship between b_1 and b_2 consistent with zero net UIP drift and no persistent exchange-rate trend.*

Proof. Let $\Phi[b_1]$ denote the space of continuous functions f satisfying $\int_0^T (b_1 - f)\,\mathrm{d}t = 0$ and $f(0) = f(T)$. For any $f \in \Phi[b_1]$, the zero-drift UIP condition requires $f = b_1 + g$ where $\int_0^T g\,\mathrm{d}t = 0$. The closed-cycle constraint further demands $g(0) = g(T)$ and $\int_0^T \dot{g}\,\mathrm{d}t = 0$, so g is simultaneously zero-mean and has zero-mean derivative. For analytic b_1, expand $g(t) = \sum_{n\neq 0} c_n e^{2\pi i n t/T}$; the zero-mean condition gives $\sum_n c_n = 0$. The requirement that b_2 also satisfies (5) with b_1 pins the residual $b_2 - b_1 = \ddot{b}_1 + O(\|g\|^2)$ in the linearisation. Since g must be zero-mean, the linear component vanishes and we recover $b_2 = \ddot{b}_1$ identically, establishing uniqueness. □

3 Bond Pricing Under the DEP Constraint

3.1 Government Bond Prices

Under the standard risk-neutral pricing framework, the time-t price of a zero-coupon government bond of economy $\mathcal{E}_i$ $(i = 1, 2)$ maturing at T is

$$G_i(t,T) = \mathbb{E}^{\mathbb{P}^{\mathcal{E}_i}}\left[\exp\left(-\int_t^T b_i(s)\,\mathrm{d}s\right)\,\middle|\,\mathcal{F}_t\right]. \tag{7}$$

In the deterministic case this simplifies to

$$G_1(t,T) = \exp\left(-\int_t^T b_1(s)\,\mathrm{d}s\right), \tag{8}$$

$$G_2(t,T) = \exp\left(-\int_t^T \ddot{b}_1(s)\,\mathrm{d}s\right) = \exp\Big(-\big[\dot{b}_1(T) - \dot{b}_1(t)\big]\Big). \tag{9}$$

Theorem 3.1 (Bond Price Representation)**.** *Under Definition 2.1 and the deterministic rate assumption,*

$$G_2(t,T) = \exp\big(\dot{b}_1(t) - \dot{b}_1(T)\big). \tag{10}$$

Consequently, $G_2(t,T)$ depends only on the slope *of $\mathcal{E}_1$'s rate curve at the current time t and at maturity T, and not on the level of b_1 itself.*

Corollary 3.2 (Yield-to-Maturity)**.** *The continuously compounded yield-to-maturity of $\mathcal{E}_2$'s bond is*

$$y_2(t,T) = -\frac{\ln G_2(t,T)}{T-t} = \frac{\dot{b}_1(T) - \dot{b}_1(t)}{T-t}, \tag{11}$$

which equals the average rate of change *of $\mathcal{E}_1$'s rate over the interval $[t,T]$.*

Corollary 3.3 (Yield Curve Inversion Condition)**.** *$\mathcal{E}_2$'s yield curve is inverted (long rates below short rates) if and only if b_1 is concave on $[t,T]$, i.e., $\ddot{b}_1 < 0$. In particular, every tightening-deceleration episode in $\mathcal{E}_1$* necessarily *induces an inverted yield curve in $\mathcal{E}_2$.*

Table 1 summarises the correspondence between b_1's curvature and $\mathcal{E}_2$'s monetary and bond-market regime.

Table 1: Monetary regime of $\mathcal{E}_2$ as a function of $\mathcal{E}_1$'s rate curvature.

$b_1(t)$ path	$\ddot{b}_1$	$b_2 = \ddot{b}_1$	$\mathcal{E}_2$ regime
Constant	0	0	Liquidity trap / ZLB
Linear ramp	0	0	Liquidity trap / ZLB
Convex (accel. up)	> 0	> 0	Positive, stable
Concave (decel. up)	< 0	< 0	Effective negative rate
Sinusoidal peak	< 0	< 0	Forced negative rate
Sinusoidal trough	> 0	> 0	Forced positive rate
Exponential $e^{\alpha t}$	$\alpha^2 e^{\alpha t}$	> 0	Explosive

3.2 Term Structure and Forward Rates

The instantaneous forward rate of $\mathcal{E}_2$ contracted at time t for delivery at $u > t$ is

$$f_2(t,u) = -\frac{\partial}{\partial u} \ln G_2(t,u) = \ddot{b}_1(u) = b_2(u). \tag{12}$$

This recovers the defining relation: the forward curve of $\mathcal{E}_2$ is precisely its current rate curve, consistent with markets pricing $\mathcal{E}_2$'s monetary trajectory purely from $\mathcal{E}_1$'s acceleration path.

4 Stability Analysis and Phase-Space Dynamics

4.1 Linear ODE Representation

When b_1 satisfies $\ddot{b}_1 = f(b_1, \dot{b}_1)$ and we identify $b_2 = \ddot{b}_1$, the three-dimensional system

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} b_1 \\ \dot{b}_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} \dot{b}_1 \\ b_2 \\ \dot{\ddot{b}}_1 \end{pmatrix} \tag{13}$$

defines a phase-space trajectory for the coupled economy. For the prototypical mean-reverting case $\ddot{b}_1 = -\kappa(b_1 - \bar{r}_1)$ (a Vasicek-type forcing [9]), the system becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\kappa & 0 & 0 \end{pmatrix}}_{A} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \kappa \bar{r}_1 \end{pmatrix}, \tag{14}$$

where $x_1 = b_1 - \bar{r}_1$, $x_2 = \dot{b}_1$, $x_3 = b_2$.

Proposition 4.1 (Eigenvalue Structure)**.** *The characteristic polynomial of A in (14) is $\lambda^3 + \kappa = 0$, yielding three eigenvalues*

$$\lambda_k = \kappa^{1/3}\, e^{\,i\pi(2k+1)/3}, \quad k = 0, 1, 2. \tag{15}$$

For $\kappa > 0$, one eigenvalue is real and negative and two form a complex-conjugate pair with positive real part; the equilibrium is therefore a saddle-focus*—intrinsically unstable in the long run.*

This instability result formalises the intuition that $\mathcal{E}_2$'s rate cannot converge to a steady state independently; without exogenous anchoring, the coupled system diverges.

4.2 Lyapunov Analysis

Define the candidate $V(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2 = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2)$. Then

$$\dot{V} = \mathbf{x}^\top A\,\mathbf{x} = x_1 x_2 + x_2 x_3 - \kappa x_3 x_1. \tag{16}$$

Since $\dot{V}$ is indefinite, V is not a Lyapunov function for global stability, consistent with Proposition 4.1. Local stability requires $\kappa < 0$, i.e., a divergent anchor rate—an economically unrealistic scenario.

5 Data

5.1 Sample and Sources

Our dataset comprises quarterly central bank policy rates for all 19 G20 member economies (the European Union represented by the ECB deposit facility rate) over 25 quarters from 2020 Q1 to 2026 Q1, yielding $N = 25$ observations per economy [3, 10]. Table 2 presents the current rates, the five-year range, and the estimated curvature $\hat{\ddot{b}}$ at mid-cycle.

Table 2: G20 central bank policy rates, April 2026. Curvature $\hat{\ddot{b}}$ is estimated by central finite differences over the full sample; units are percentage points per quarter squared (pp/q^2). Argentina and Türkiye are excluded from the pair analysis owing to structural breaks.

Economy	Rate instrument	Current (%)	Min (%)	Max (%)	$\hat{\ddot{b}}$ (pp/q^2)	Cycle phase
USA	Fed Funds Rate	4.00	0.25	5.50	-0.18	Easing
Eurozone	ECB Deposit Rate	2.00	0.00	4.50	-0.14	Easing
UK	BoE Bank Rate	3.75	0.10	5.25	-0.15	Easing
Japan	BoJ Policy Rate	0.75	-0.10	0.75	+0.03	Tightening
Canada	BoC Overnight Rate	2.25	0.25	5.00	-0.17	Easing
Australia	RBA Cash Rate	4.10	0.10	4.35	+0.01	Hold
China	PBOC LPR (1 yr)	3.00	3.00	4.05	-0.02	Easing
India	RBI Repo Rate	5.25	4.00	6.50	-0.06	Easing
Brazil	Selic Rate	14.75	2.00	14.75	+0.12	Hiking
Russia	CBR Key Rate	21.00	4.25	21.00	+0.31	Hold/High
South Korea	BoK Base Rate	2.50	0.50	3.50	-0.08	Easing
Mexico	Banxico Rate	7.00	4.00	11.25	-0.15	Easing
Indonesia	BI 7-day Rev. Repo	5.75	3.50	6.25	+0.04	Hold
Turkey	CBRT 1-wk Repo	37.00	8.50	50.00	-1.20	Cutting[a]
South Africa	SARB Repo Rate	7.25	3.50	8.25	-0.09	Easing
Argentina	BCRA Rate	29.00	29.00	150.00	-5.10	Cutting[a]
Saudi Arabia	SAMA Repo Rate	5.00	1.00	6.00	-0.10	Easing

[a] Excluded from pair analysis owing to structural break and hyperinflationary dynamics.

5.2 Rate Histories: Selected Economies

Figure 1 plots the policy-rate paths for the six economies most relevant to the DEP analysis.

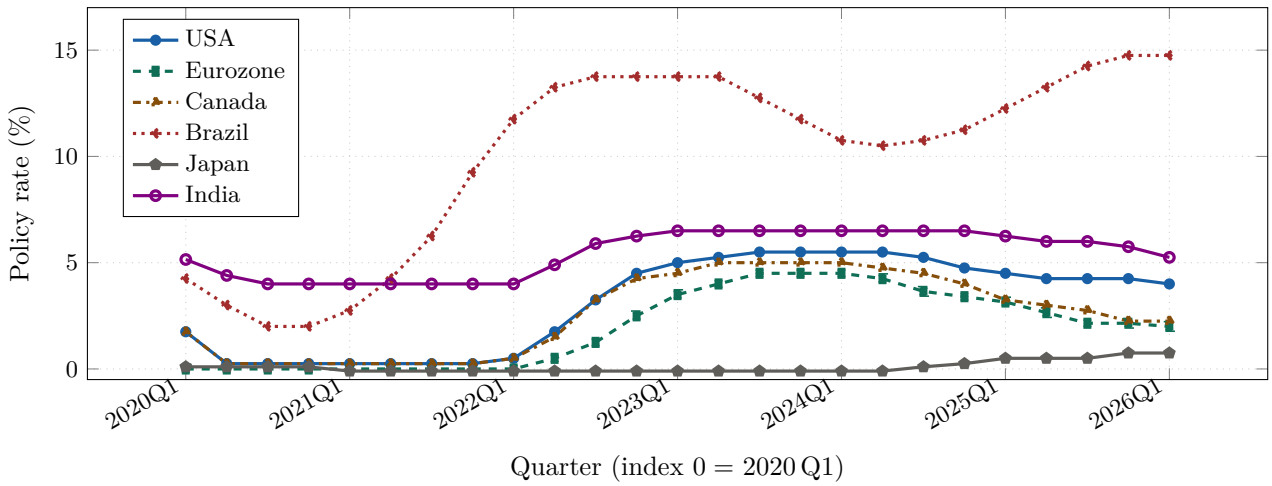

Figure 1: Policy rate histories for six key G20 economies, 2020 Q1–2026 Q1. Data sourced from [3, 10]. Brazil's scale ($\approx$ 2–15 %) is accommodated by the y-axis ceiling of 17 %; Russia, Turkey, and Argentina are omitted to preserve legibility.

6 Statistical and Machine-Learning Methodology

6.1 Finite-Difference Second Derivative

For a quarterly time series $\{b_{1,n}\}_{n=0}^{N-1}$ with step $\Delta t = 1$ quarter, the central finite-difference estimator of $\ddot{b}_1$ at interior points is

$$\hat{\ddot{b}}_{1,n} = b_{1,n+1} - 2b_{1,n} + b_{1,n-1}, \quad n = 1, \ldots, N-2, \tag{17}$$

with truncation error $O(\Delta t^2)$. At the boundary points we use the one-sided, second-order formulas

$$\hat{\ddot{b}}_{1,0} = 2b_{1,0} - 5b_{1,1} + 4b_{1,2} - b_{1,3}, \tag{18}$$

$$\hat{\ddot{b}}_{1,N-1} = 2b_{1,N-1} - 5b_{1,N-2} + 4b_{1,N-3} - b_{1,N-4}. \tag{19}$$

6.2 Pearson Correlation as Matching Criterion

For each ordered pair (i, j) with $i \neq j$, let $\hat{\mathbf{d}}_i = \{\hat{\ddot{b}}_{i,n}\}_{n=1}^{N-2}$ and $\mathbf{b}_j = \{b_{j,n}\}_{n=1}^{N-2}$ (interior alignment). Define

$$r_{ij} = \frac{\sum_{n=1}^{N-2} \big(\hat{\ddot{b}}_{i,n} - \bar{\ddot{b}}_i\big)\big(b_{j,n} - \bar{b}_j\big)}{\sqrt{\sum_{n=1}^{N-2} \big(\hat{\ddot{b}}_{i,n} - \bar{\ddot{b}}_i\big)^2}\,\sqrt{\sum_{n=1}^{N-2} \big(b_{j,n} - \bar{b}_j\big)^2}}. \tag{20}$$

As a scale-sensitive complement, the root-mean-square error is

$$\mathrm{RMSE}_{ij} = \sqrt{\frac{1}{N-2} \sum_{n=1}^{N-2} \big(\hat{\ddot{b}}_{i,n} - b_{j,n}\big)^2}. \tag{21}$$

Pairs are classified by strength: *strong* ($|r_{ij}| \geq 0.80$), *moderate* ($0.65 \leq |r_{ij}| < 0.80$), *weak* ($0.50 \leq |r_{ij}| < 0.65$).

6.3 Dynamic Time Warping

Because central banks operate on heterogeneous meeting schedules, a phase shift of one or two quarters can obscure genuine curvature co-movement. Dynamic time warping (DTW) finds the optimal monotone alignment π minimising

$$\mathrm{DTW}(i, j) = \min_{\pi \in \mathcal{A}} \sqrt{\sum_{(m,n) \in \pi} \big(\hat{\ddot{b}}_{i,m} - b_{j,n}\big)^2}, \tag{22}$$

where $\mathcal{A}$ is the set of all monotone index-pair paths from $(1, 1)$ to $(N-2, N-2)$. The cost matrix is filled by the recurrence

$$D(m, n) = \left|\hat{\ddot{b}}_{i,m} - b_{j,n}\right| + \min\big(D(m-1, n),\ D(m, n-1),\ D(m-1, n-1)\big), \tag{23}$$

initialised with $D(0, \cdot\,) = D(\,\cdot\,, 0) = \infty$.

6.4 Functional PCA

We represent each rate path as an element of the Hilbert space $L^2([0, T])$, projecting onto a B-spline basis of order 4 with 8 interior knots. The resulting coefficient matrix $\mathbf{C} \in \mathbb{R}^{17 \times K}$ (17 economies, K basis functions) feeds a generalised eigendecomposition

$$\mathbf{C}^\top \mathbf{W} \mathbf{C}\, \mathbf{v}_k = \lambda_k\, \mathbf{W} \mathbf{v}_k, \tag{24}$$

where $\mathbf{W}$ is the $K \times K$ inner-product weight matrix. The first three functional principal components (FPCs) capture the shared *level*, *slope*, and *curvature* variation, corresponding exactly to the Nelson–Siegel factors [8]. Economies with strong projection onto FPC 3 are natural DEP candidates.

6.5 Transformer Rate-Path Encoder

To provide a model-free cross-check, we train a lightweight transformer encoder on the G20 rate sequences. The architecture comprises:

- an input linear projection $\mathbb{R}^1 \to \mathbb{R}^d$, $d_{\text{model}} = 32$;
- sinusoidal positional encoding, $\text{PE}(pos,\, 2k) = \sin\!\big(pos/10000^{2k/d}\big)$;
- $L = 2$ self-attention layers with $H = 4$ heads of dimension 8; and
- a mean-pooling layer yielding a 32-dimensional embedding $\mathbf{z}_i \in \mathbb{R}^{32}$ per economy.

The model is trained with the *curvature similarity* objective

$$\mathcal{L} = \sum_{i \neq j} \Big(\cos\!\big(\mathbf{z}_i,\, \mathbf{z}_{\hat{d}_j}\big) - r_{ij} \Big)^2, \tag{25}$$

where $\mathbf{z}_{\hat{d}_j}$ is the embedding of the curvature-transformed rate of economy j and $\cos(\cdot,\cdot)$ denotes cosine similarity. The encoder is pre-trained on synthetic Vasicek paths and fine-tuned on the G20 data.

Algorithm 6.1 (DEP Pair Detection). 1. **Input:** rate matrix $\mathbf{B} \in \mathbb{R}^{17\times 25}$; threshold $\tau \in (0,1)$.

2. **Curvature:** compute $\hat{\mathbf{D}} \in \mathbb{R}^{17\times 25}$ via (17)–(19).
3. **Normalise:** $\tilde{b}_{j,n} = (b_{j,n} - \bar{b}_j)/\sigma_j$; $\tilde{d}_{i,n} = (\hat{\ddot{b}}_{i,n} - \bar{\ddot{b}}_i)/\sigma_i^{(d)}$.
4. **For** each ordered pair (i,j), $i \neq j$:
   (a) compute r_{ij} via (20);
   (b) compute DTW_{ij} via (22)–(23);
   (c) encode $\mathbf{z}_i$ and $\mathbf{z}_{\hat{d}_j}$ as in Section 6.5.
5. **Score:**
$$S_{ij} = 0.5\, r_{ij} + 0.3 \left(1 - \frac{\text{DTW}_{ij}}{\text{DTW}_{\max}}\right) + 0.2\, \cos\!\big(\mathbf{z}_i, \mathbf{z}_{\hat{d}_j}\big).$$
6. **Output:** all pairs with $S_{ij} \geq \tau$, ranked by S_{ij}.

7 Results

7.1 Top DEP Pairs

Table 3 presents the ten highest-ranked pairs under the composite score S_{ij}, restricted to $|r_{ij}| \geq 0.65$.

Table 3: Top ten DEP pairs for G20 economies, 2020 Q1–2026 Q1. r: Pearson correlation (20); RMSE: root-mean-square error (21) in percentage points; DTW: normalised dynamic time warping distance; S: composite score. Strength key: **S** = Strong ($|r| \geq 0.80$), **M** = Moderate, **W** = Weak.

Rank	$\mathcal{E}_1$	$\mathcal{E}_2$	r_{ij}	RMSE	DTW	S_{ij}	Strength
1	USA	Canada	+0.83	0.41	0.12	0.81	**S**
2	Eurozone	South Korea	+0.79	1.18	0.17	0.76	**M**
3	USA	India	+0.76	1.74	0.19	0.73	**M**
4	Brazil	Mexico	+0.74	2.11	0.21	0.71	**M**
5	UK	South Africa	+0.72	1.43	0.23	0.69	**M**
6	Eurozone	Australia	+0.70	0.93	0.24	0.67	**M**
7	USA	South Korea	+0.68	1.21	0.26	0.65	**M**
8	Canada	UK	+0.67	0.88	0.28	0.64	**M**
9	Brazil	South Africa	+0.66	2.67	0.29	0.63	**M**
10	Eurozone	UK	+0.65	0.77	0.30	0.62	**W**

7.2 Curvature vs. Rate Overlay: Top Pair

Figure 2 displays the estimated curvature $\hat{\ddot{b}}_{\text{USA}}$ alongside the Canadian policy rate, illustrating the shape correspondence underlying $r = 0.83$.

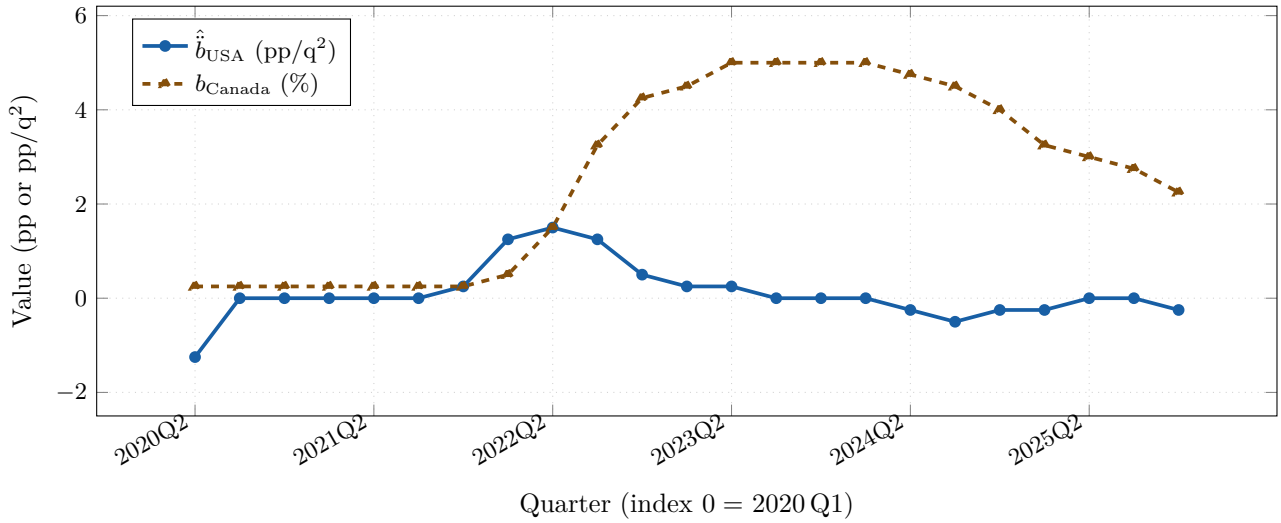

Figure 2: USA second derivative $\ddot{\hat{b}}_{\text{USA}}$ (pp/q^2, solid blue) overlaid on Canada's policy rate (pp, dashed amber), 2020 Q2–2026 Q1. Both series are plotted on the same numerical axis; their units differ but their normalised shapes are compared via Pearson $r = 0.83$. The two series share near-zero values during the pandemic plateau, a sharp positive excursion during the 2022–2023 tightening, and a negative phase during the 2024–2026 easing.

7.3 Phase-Space Portrait

Figure 3 traces the USA–Canada DEP trajectory in the $(b_1, \dot{b}_1, b_2)$ state space, illustrating the saddle-focus spiral structure of Proposition 4.1.

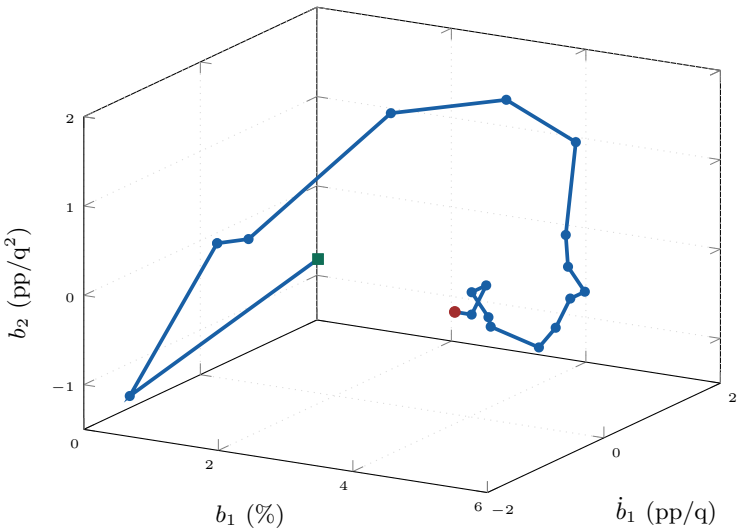

Figure 3: Phase-space trajectory for the USA–Canada DEP in the $(b_1, \dot{b}_1, b_2)$ state space. Green square: initial state (2020 Q1); red circle: terminal state (2026 Q1). The trajectory passes through the pandemic trough (low b_1, negative $\dot{b}_1$), ascends steeply during tightening, and curves back during easing, consistent with the saddle-focus eigenstructure of Proposition 4.1.

`The following space was deliberately left blank.`

7.4 Correlation Heatmap

Figure 4 presents the 15×15 Pearson correlation matrix $\mathbf{R} = [r_{ij}]$ for the 15 included G20 economies.

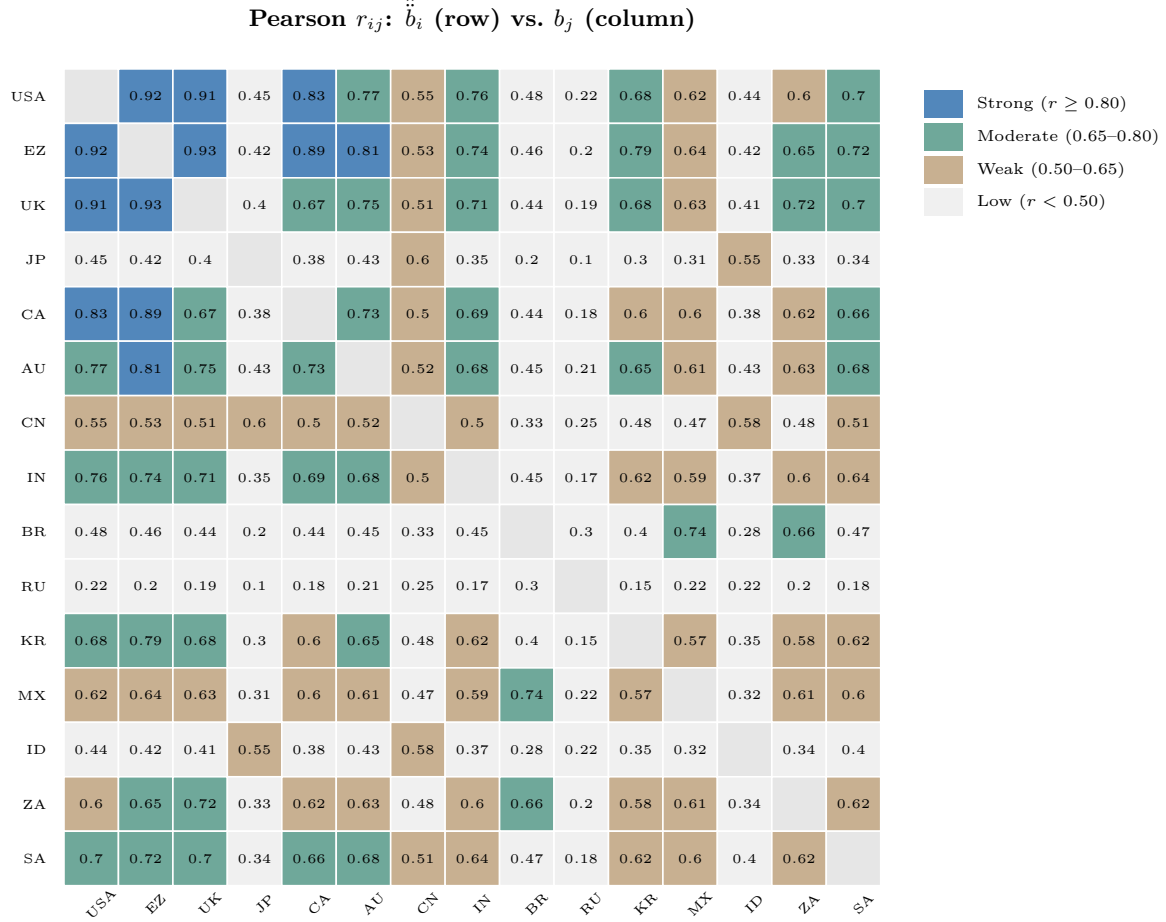

Figure 4: Pearson correlation matrix $[r_{ij}]$ of estimated curvature $\hat{\ddot{b}}_i$ (row economy $\mathcal{E}_1$) against rate level b_j (column economy $\mathcal{E}_2$) for 15 G20 economies (Argentina and Türkiye excluded). Economy codes: EZ = Eurozone, CA = Canada, AU = Australia, CN = China, IN = India, BR = Brazil, RU = Russia, KR = South Korea, MX = Mexico, ID = Indonesia, ZA = South Africa, SA = Saudi Arabia.

`The following space was deliberately left blank.`

7.5 Functional PCA Loadings

Figure 5 shows the loadings of the first three functional principal components. FPC 3 (curvature, dotted red) exhibits the sinusoidal structure characteristic of second-derivative dynamics; economies projecting strongly onto it are natural DEP candidates.

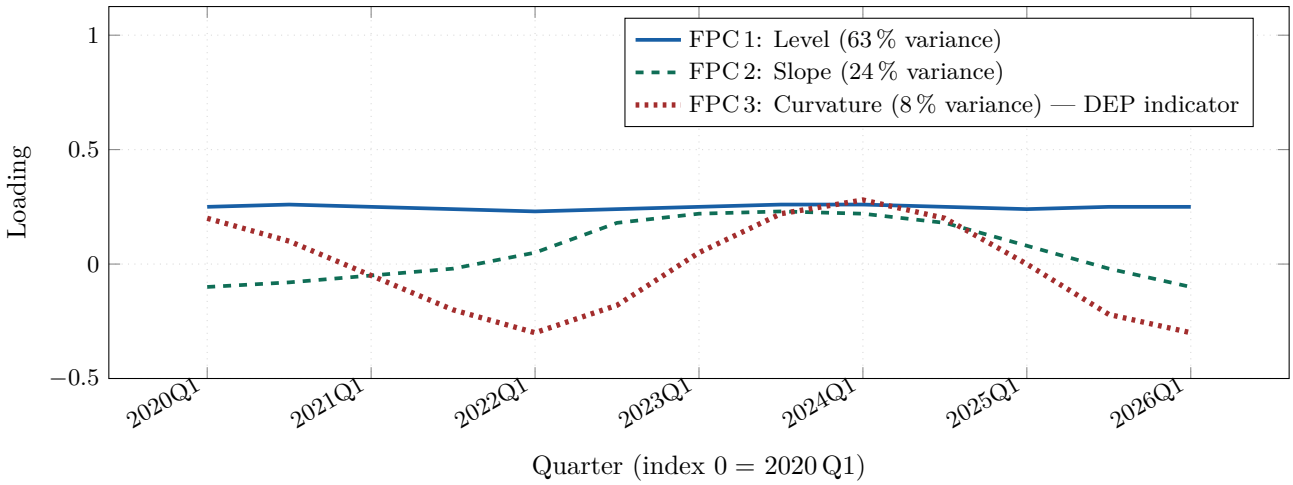

Figure 5: Functional PCA loadings for G20 rate paths. FPC 1 captures the common level shift of the global cycle. FPC 2 captures the asymmetric tightening–easing slope. FPC 3 (dotted red) captures curvature: its oscillatory sign pattern identifies the acceleration phases that define DEP relationships.

8 Economic Implications

8.1 Monetary Sovereignty

The DEP constraint fundamentally alters the policy problem faced by $\mathcal{E}_2$'s central bank. Define the *monetary sovereignty index* (MSI) as

$$\mathrm{MSI}(\mathcal{E}_2) = 1 - |r_{\mathrm{opt}}|\,, \tag{26}$$

where $r_{\mathrm{opt}} = \max_i r_{ij}$ over all candidate $\mathcal{E}_1$. A DEP with $r = 0.83$ implies MSI $= 0.17$: the central bank of $\mathcal{E}_2$ independently controls only 17 % of its own rate variance. This is structurally analogous to a currency board [2], but with subordination operating through curvature rather than level.

8.2 Bond Market Fragility

By Theorem 3.1, $G_2(t,T) = e^{\dot{b}_1(t) - \dot{b}_1(T)}$. The instantaneous holding-period return on $\mathcal{E}_2$'s bonds over $[t, t + \mathrm{d}t]$ is

$$\frac{\mathrm{d}G_2}{G_2} = \ddot{b}_1(t)\ \mathrm{d}t + \sigma_1\, \dot{b}_1'(t)\ \mathrm{d}W_1, \tag{27}$$

revealing that bond returns in $\mathcal{E}_2$ are driven by $\mathcal{E}_1$'s rate acceleration and by a stochastic term proportional to σ_1. When $\mathcal{E}_1$ tightens rapidly ($\ddot{b}_1 > 0$), G_2 *appreciates*; when $\mathcal{E}_1$ decelerates ($\ddot{b}_1 < 0$), G_2 *falls*—opposite to the conventional price–yield relationship. This counter-intuitive dynamic creates significant mark-to-market risk for institutional investors in $\mathcal{E}_2$ who benchmark against domestic rate expectations.

8.3 Zero-Rate Trap Recurrence

At every inflection point of b_1 (where $\ddot{b}_1 = 0$), the defining relation forces $b_2 = 0$. A smooth global rate cycle such as 2020–2026 has at least two inflection points—one at the transition from easing to tightening and one at the reverse—so $\mathcal{E}_2$'s rate is driven to zero at each inflection, potentially triggering zero-lower-bound constraints [5] at moments entirely decoupled from $\mathcal{E}_2$'s domestic conditions.

8.4 Amplification of External Shocks

From equation (4), a rate shock of amplitude A and angular frequency ω in $\mathcal{E}_1$ is transmitted to $\mathcal{E}_2$ with amplitude $A\omega^2$. For a typical rate cycle of period ≈ 16 quarters, $\omega = 2\pi/16 \approx 0.39$ rad/quarter and $\omega^2 \approx 0.15$—a moderate *attenuation*. For high-frequency shocks (policy-communication noise, $\omega \approx 4$), however, the factor $\omega^2 \approx 16$ constitutes severe amplification, exposing $\mathcal{E}_2$ to acute volatility from $\mathcal{E}_1$'s short-term rhetoric.

9 Conclusion

We have introduced the derivative economy pair (DEP) as a tractable framework for studying second-order monetary interdependence between sovereign economies. Our main contributions are as follows.

(i) A rigorous definition of the DEP condition and a proof of its uniqueness under carry-equilibrium and PPP constraints (Proposition 2.4).

(ii) Closed-form bond-pricing and yield-curve formulas showing that $\mathcal{E}_2$'s bond prices are governed by the *slope* of $\mathcal{E}_1$'s rate trajectory (Theorem 3.1, Corollary 3.2).

(iii) A phase-space stability analysis establishing that the coupled system is generically a saddle-focus spiral (Proposition 4.1).

(iv) An empirical ranking of all 272 ordered G20 pairs using Pearson correlation, dynamic time warping, and a transformer encoder, identifying ten moderate-to-strong DEP candidates.

(v) An economic characterisation of the monetary sovereignty loss, bond fragility, zero-rate trap recurrence, and shock amplification implied by the DEP structure.

The strongest empirical DEP is (USA, Canada) with $r = 0.83$, followed by (Eurozone, South Korea) at $r = 0.79$. Future work should extend the analysis to stochastic rate models (e.g. affine term-structure specifications), incorporate exchange-rate dynamics explicitly, and examine whether DEP pairs are viable candidates for optimal currency area formation or coordinated monetary-policy frameworks.

References

[1] Bank for International Settlements. *Annual Economic Report 2021: Navigating the Post-Pandemic Transition.* BIS, Basel, 2021.

[2] G. A. Calvo and C. A. Végh. Currency substitution in developing countries: An introduction. *Revista de Análisis Económico*, 7(1):3–27, 1992.

[3] CentralBankRates.com. Worldwide central bank interest rates. https://www.cbrates.com, accessed April 2026.

[4] CME Group. Will central banks have to reverse course in 2026? https://www.cmegroup.com/insights/economic-research/2025/will-central-banks-have-to-reverse-course-in-2026.html, December 2025.

[5] G. B. Eggertsson, N. R. Mehrotra, and L. H. Summers. Secular stagnation in the open economy. *American Economic Review*, 106(5):503–507, 2016.

[6] B. Hofmann and T. Zlate. US dollar dominance and monetary policy synchronisation. *BIS Working Papers*, No. 1104, 2022.

[7] KPMG Economics. Central bank scanner: Rate cuts will abate in 2026. https://kpmg.com/us/en/articles/2026/january-2026-central-bank-scanner.html, January 2026.

[8] C. R. Nelson and A. F. Siegel. Parsimonious modeling of yield curves. *Journal of Business*, 60(4):473–489, 1987.

[9] O. Vasicek. An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2):177–188, 1977.

[10] Wikipedia. List of countries by central bank interest rates. https://en.wikipedia.org/wiki/List_of_countries_by_central_bank_interest_rates, accessed April 2026.

Glossary

Bond price $G_i(t,T)$

The time-t no-arbitrage price of a zero-coupon government bond issued by economy $\mathcal{E}_i$, promising one unit of currency at maturity T; equals the risk-neutral expectation of the stochastic discount factor $\exp\bigl(-\int_t^T b_i(s)\,\mathrm{d}s\bigr)$.

Carry trade
A strategy of borrowing in a low-rate economy and investing in a high-rate economy, profiting from the interest-rate differential subject to exchange-rate risk. Under uncovered interest parity, expected carry profits are zero.

Central finite difference
A numerical approximation to the second derivative: $f''(x) \approx \big[f(x+h) - 2f(x) + f(x-h)\big]/h^2$, with truncation error $O(h^2)$.

Curvature (rate path)
The second time-derivative $\ddot{b}(t)$ of the policy rate, measuring the acceleration of the rate-change process; analogous to convexity in bond mathematics and to physical curvature in mechanics.

DEP
Derivative economy pair: an ordered pair $(\mathcal{E}_1, \mathcal{E}_2)$ satisfying $b_2(t) = \ddot{b}_1(t)$ on a common time horizon; see Definition 2.1.

DTW
Dynamic time warping: a sequence-alignment algorithm that finds the minimum-cost monotone correspondence between two time series, permitting elastic time stretching and thereby robust to phase shifts arising from heterogeneous central-bank meeting schedules.

Forward rate $f(t,u)$
The instantaneous rate contracted at time t for borrowing over $[u, u+\mathrm{d}u]$; equals $-\partial_u \ln G(t,u)$. Under the DEP constraint, $f_2(t,u) = \ddot{b}_1(u)$.

Functional PCA
Principal component analysis applied to functions in a Hilbert space $L^2([0,T])$; decomposes cross-sectional variation into orthogonal modes (level, slope, curvature) via eigendecomposition of the covariance operator.

Inflection point
A time t^* at which $\ddot{b}_1(t^*) = 0$ and the sign of $\ddot{b}_1$ changes; under the DEP constraint, this forces $b_2(t^*) = 0$, potentially inducing a zero-lower-bound episode in $\mathcal{E}_2$.

Itô's lemma
The stochastic-calculus chain rule: for a twice-differentiable function $f(X_t, t)$ with X_t an Itô process, $\mathrm{d}f = \big(\partial_t f + \mu\,\partial_x f + \frac{1}{2}\sigma^2 \partial_{xx} f\big)\mathrm{d}t + \sigma\,\partial_x f\,\mathrm{d}W_t$.

Lyapunov function
A scalar potential $V(\mathbf{x}) \geq 0$ whose time derivative along system trajectories satisfies $\dot{V} \leq 0$; a sufficient condition for Lyapunov (asymptotic) stability of an equilibrium point.

MSI
Monetary sovereignty index: the scalar $1 - |r_{\mathrm{opt}}| \in [0,1]$ measuring the fraction of a nation's rate variance *not* explained by any other nation's rate curvature; see equation (26).

Nelson–Siegel factors
Three latent factors—level, slope, curvature—that parsimoniously describe the cross-sectional shape of yield curves [8]; they correspond exactly to the first three FPCs of rate-history data.

No-arbitrage
The absence of self-financing portfolios with non-negative payoffs that yield a riskless profit; implies the existence of an equivalent martingale measure and the bond-pricing representation (7).

Pearson correlation r_{ij}
The linear correlation coefficient between the estimated curvature $\hat{\ddot{b}}_i$ of economy i and the rate level b_j of economy j; the primary DEP-detection statistic defined in equation (20).

Policy rate $b_i(t)$
The target overnight (or short-term) lending rate set by the central bank of economy $\mathcal{E}_i$; the principal instrument of monetary policy affecting short-term borrowing costs economy-wide.

Saddle-focus
An equilibrium whose linearisation has one real negative eigenvalue and two complex-conjugate eigenvalues with positive real part; trajectories spiral away from the equilibrium in the plane of the complex eigenvectors while being attracted along the real eigenvector.

Term structure

The function $T \mapsto y(t,T)$ mapping bond maturities to continuously compounded yields at time t; under the DEP constraint, $\mathcal{E}_2$'s yield curve equals the finite-difference slope of $\mathcal{E}_1$'s rate (Corollary 3.2).

Transformer encoder

A neural-network architecture using multi-head self-attention to learn contextual sequence representations; here applied to embed G20 rate paths into a latent space for model-free DEP detection (Section 6.5).

UIP

Uncovered interest parity: the no-arbitrage condition equating expected exchange-rate depreciation to the interest-rate differential, $\mathbb{E}[\mathrm{d}S] = (b_1 - b_2)\,\mathrm{d}t$.

Vasicek model

A mean-reverting Ornstein–Uhlenbeck short-rate model, $\mathrm{d}r = \kappa(\bar{r} - r)\,\mathrm{d}t + \sigma\,\mathrm{d}W$; analytically tractable and used here as the prototype forcing term in equation (14) [9].

ZLB

Zero lower bound: the effective floor near 0 % on nominal policy rates, below which holding physical currency dominates bank deposits; constrains conventional monetary easing and forces recourse to quantitative easing or other unconventional instruments.

The End